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SPCE 5085
Assignment #2

4.1 — Calculate the path loss for a satellite to earth down link with a distance of 38,500 km at a frequency of 4 GHz. Whaat is the uplink path loss if the earth station transmits to the same satellite at a frequency of 6.0 GHz?

$$L_p \text{ [dB]} = 20\log_{10}\left(\frac{4\pi R}{\lambda}\right), \text{ where } \lambda = c/f$$

$$L_p(4\ GHz) = 20\log_{10}\left(\frac{4\pi \times 38{,}500{,}000\ m}{299{,}792{,}458/4 \times 10^9\ m}\right) = \boxed{196.20\ dB}$$

$$L_p(6\ GHz) = 20\log_{10}\left(\frac{4\pi \times 38{,}500{,}000\ m}{299{,}792{,}458/6 \times 10^9\ m}\right) = \boxed{199.72\ dB}$$

4.2 — A LEO satellite at an altitude of 400 km has a distance to a receiving terminal of 500 km. The satellite radiates a power of 1.0 W from a spot beam antenna with a gain of 35 dB in the direction of the terminal. Find the flux density at the receiving terminal, and the power received with a tracking antenna with a gain of 30 dB. (Assuming an operating frequency of 4 GHz.)

$$F = \frac{P_t G_t}{4\pi R^2}, \text{ where } G_t = 10^{35/10} \text{ (linear) and } R = 500{,}000\ m$$

$$F = \frac{1\ W \times 10^{35/10}}{4\pi(500{,}000\ m)^2} = 1.007 \times 10^{-9}\ W/m^2 = \boxed{-89.97\ dBW/m^2}$$

$$\text{EIRP [dBW]} = 10\log_{10}(P_t) + G_t\text{[dB]} = 0\ dBW + 35\ dB = 35\ dBW$$

$$P_r \text{ [dBW]} = \text{EIRP} + G_r - L_p = 35\ dBW + 30\ dB - 158.47\ dB = \boxed{-93.47\ dBW = 4.50 \times 10^{-10}\ W}$$

4.4 — A Ku-band DBS-TV receiving terminal has an antenna with a gain of 34 dB at 12.5 GHz. The noise temperature of the receiver under clear air conditions is 80 K.

a Calculate the G/T ratio of the terminal in dB.

$$G/T \text{ [dB/K]} = G\text{[dB]} - 10\log_{10}(T)$$

$$G/T = 34\ dB - 10\log_{10}(80\ K) = 34 - 19.03 = \boxed{14.97\ dB/K}$$

4.8 — A Ku-band satellite carries a bent pipe transponder with 36 MHz bandwidth, which is operated in its linear region. In the clear air conditions, the uplink CNR is 24.0 dB and the downlink CNR at a receiving earth station is 14.0 dB.

a Calculate the overall $(\text{CNR})_o$ in clear sky conditions.

$$\text{CNR}_o \text{ [dB]} = -10\log_{10}\left(10^{-\text{CNR}_u\text{[dB]}/10} + 10^{-\text{CNR}_d\text{[dB]}/10}\right)$$

$$\text{CNR}_o = -10\log_{10}\left(10^{-24.0/10} + 10^{-14.0/10}\right) = \boxed{13.59\ dB}$$

Source:
https://github.com/chrismcarrico/homework/blob/master/spce5085/a2/carrico_spce5085_a2.ipynb

b Rain affects the uplink causing 4 dB of attenuation. Calculate the overall $(\text{CNR})_o$ at the receiving earth station.

$\text{CNR}_u[\text{dB}] = 24.0\ dB - 4\ dB = 20.0\ dB$; CNR_d is unchanged at 14.0 dB.

$\text{CNR}_o = -10\log_{10}\left(10^{-20.0/10} + 10^{-14.0/10}\right) = \boxed{13.03\ dB}$

4.11 — Calculate the power received by the earth station when the satellite in Example 4.1 has a regional beam with a gain of 26 dB.

Not solved — this part depends on Example 4.1's transmit power/EIRP, distance, and global beam gain, which were not available.

i What is the diameter of a circular aperture antenna with a gain of 26 dB at 11 GHz?

$G = \eta_A \left(\frac{\pi D}{\lambda}\right)^2 \Rightarrow D = \frac{\lambda}{\pi}\sqrt{\frac{G}{\eta_A}}$

$\lambda = c/f = 0.02725\ m$

$G = 10^{26/10} = 398.11$ (linear)

$D = \frac{0.02725\ m}{\pi}\sqrt{\frac{398.11}{0.55}} = \boxed{0.233\ m}$